Jeet Bahadur Thapa
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Dayton, Ohio | 937-414-7125 (Open to Relocate)

EDUCATION:

- **Master of Science in Mechanical Engineering** — Jan 2024 – Present
  Wright State University, Dayton, Ohio — Cumulative GPA: 3.857 / 4.0
- **Bachelor of Science in Mechanical Engineering** — Dec 2016
  Tribhuvan University, Kathmandu, Nepal — Overall GPA: 3.0 / 4.0

RELEVANT COURSEWORK:

- Advanced Thermodynamics
- Engineering Vibration & Advance Dynamics
- Advanced Mechanics of Solids
- Automotive Technology
- Finite Element Method/Analysis
- Additive Manufacturing & Machine Design

SOFTWARE AND TECHNICAL SKILLS:

- *CAD/Simulation*: SolidWorks, Simufact, ANSYS
- *Data Analysis*: MATLAB, Python
- *Testing Equipment*: Instron (UTS tensile test, 3-point bending test, Impact test, Fatigue test) MTS (ASTM D695 compression test), Keyence VR6000 (surface roughness), Mitutoyo Vision System (dimensional accuracy), Scanning Electron Microscope (SEM), X- ray CT scanner
- *3D Printing*: Formlabs Fuse 1 (Selective Laser Sintering - SLS), Fused Deposition Modeling (FDM), Preform, Prusca Slicer
- *Vibration Analysis*: Spectral Dynamics Bobcat analyzer, RMS/time-domain/frequency-domain analysis

CERTIFICATIONS, AWARDS & RESEARCH PRESENTATIONS:

- Certification of SolidWorks Design Professional (CSWP & CSWP Certified) — 2025
- Certification of SolidWorks Associate (CSWA) for 3D Design — 2024
- Certification of SolidWorks Additive Manufacturing Associate (CSWA-AM) — 2024
- Certification of SolidWorks Drawing Tools Professional (CSWPA-DT) — 2024
- OASIS RAPID Certification on Semiconductors & Integrated Manufacturing — 2024
- **Poster Presenter, MS&T 2025 Conference & ASME Dayton Engineering & Science Symposium (DESS) 2025** : Presented research on *"Environmental Effects on Mechanical Behavior of 3D-Printed Glass Fiber-Reinforced PA12 Composites,* demonstrating how build orientation, heat-treatment, & moisture alter surface integrity and tensile performance in SLS-fabricated specimens.

PROFESSIONAL EXPERIENCE:

Graduate Research Assistant (GRA)

Wright State University, Dayton, Ohio, USA
May 2025 – July 2025 & *Jan 2025 – Present*

Additive Manufacturing with Mechanical and Physical Characterization

- *SLS Printing (Formlabs Fuse 1)* – Printed PA12 and PA12-GF specimens in multiple build orientations for ASTM D638 tensile, ASTM D695 compression, & ISO 179-1 Charpy impact sample sets
- *Environmental conditioning (tensile specimens):* Conditioned ASTM D638 samples in Watlow F4T chamber and applied controlled heat treatment to compare moisture/heat effects on printed parts.
- *Mitutoyo Vision System + Keyence VR6000:* Verified specimen dimensions and measured surface roughness/topography (Sa/Sq) to correlate dimensional accuracy + surface quality with orientation/conditioning effects and explain changes in mechanical performance.
- *Compression Testing (MTS 244.51)* – Performed ASTM D695 compression tests at 0.01, 1, 3, and 5 s^{-1} to evaluate strain-rate and build-orientation effects on modulus, yield, and stress–strain behavior.
- *Impact Testing (Instron Ceast 9050)* – Conducted Charpy impact testing to compare orientation-dependent toughness and energy absorption of printed specimens.
- *FEA (ANSYS/Simufact):* Built compression models to visualize stress/deformation and validate results against ASTM D695 tests.

Material Characterization (Assisted; under faculty/lab specialist supervision for all these below)

- *XRD (Rigaku SmartLab*): Assisted θ–2θ scans on PA12 and PA12-GF powders and printed parts; prepared specimens and compiled diffractograms to compare phases/crystallinity before vs after SLS printing

- *DSC:* Extracted parameters for melting/crystallization behavior (Tm, Tc, ΔHm) to establish baseline thermal transitions and crystallinity for Fuse 1 powders parts where supplier data is limited.
- *TGA:* Extracted mass-loss curves & degradation onset to benchmark thermal stability for neat PA12 vs PA12-GF.

Graduate Teaching Assistant (GTA) Wright State University, Dayton, Ohio, USA

Mechanical Vibrations Lab: Aug 2025 – Dec 2025

- *System Response & Modal Analysis:* Conducted SDOF vibration experiments including cantilever beam and forced vibration setups to evaluate stiffness, damping, natural frequency, and mode shapes.
- *FRF & Transmissibility Analysis* – Guided frequency response and transmissibility analysis using accelerometers, impulse hammers, and rotating unbalance excitation, linking experimental results with vibration theory.

Mechanical and Material Testing Lab: Jan 2025 – April 2025

- *Tensile Test* – Demonstrated material strength testing using Instron.
- *Elastic Beam Bending*– Assisted in 3-point & 4-point bending test to observe flexural behavior.
- *Heat Treatment* – Explained effects on microstructure and hardness.
- *Impact Test* – Demonstrated Charpy testing for fracture energy.
- *Temperature Effect* – Showed how elevated temperatures affect creep and material behavior.
- *Fatigue Test* – Assisted in testing endurance limit and cycle failure.

Technical Advisor: B & K Enterprises, Bhaktapur, Nepal Dec 2021 – Nov 2023

- *Vehicle Repair SOP Implementation:* Implemented advanced service SOPs across multiple vehicle classes, improving workflow efficiency and saving 225+ labor hours/year.
- *Failure Analysis & Performance Review:* Led component failure investigations (hydraulics, clutches, brake systems), implemented preventive actions reducing repeat failures by ~15%, and evaluated technical reports to guide engineer performance and promotions

Sr. Service Engineer: B & K Enterprises, Bhaktapur, Nepal Dec 2018 – Dec 2021

- *Technical Reporting & Coordination:* Prepared warranty, and inventory reports using Microsoft Office and Microsoft Dynamics Navision, and coordinating team activities to meet delivery, cost, and quality targets.
- *Emission Compliance Inspection:* Evaluated vehicle exhaust systems for regulatory compliance using exhaust gas analyzers (petrol) and opacity meters (diesel)

Service Advisor: Sipradi Trading Pvt. Ltd. Kathmandu, Nepal Jul 2016 – Dec 2018

- *Process Improvement & Training**:*** Improved diagnostics using TATA Diagnostics Software**,** increasing service turnover by 20%**,** and trained technicians after TATA Motors HQ programs**.**
- *Advanced Alignment Technology**:*** Implemented Hunter Hawkeye Elite 3D alignment for improved accuracy and customer satisfaction.

Service Engineering Trainee: Sipradi Trading Pvt. Ltd., Kathmandu, Nepal Apr 2016 – Jul 2016

- *Fault Diagnosis* – Trained to spot mechanical & electronic issue with sensor data, circuit diagram & code scanner
- *Hands-on Repair* – Practiced disassembly/reassembly of engines, transmissions, and crown units.
- *Performance* – Achieved top training results, earning promotion to Service Advisor at a new branch

Mechanical Engineering Trainee (: CG Foods Nepal, Nawalparasi, Nepal Oct 2014 – Nov 2014

- *Machinery Operation & Maintenance*: Operated extruders, boilers, and conveyors; performed preventive maintenance, welding, lathe machining, and alignment.
- *Quality Assurance:* Calibrated equipment and ensured material consistency in compliance with plant standards

PROJECTS:

Retrofitting of a 2-stroke Gasoline Engine to a Compressed Air Engine Nov 2014 – Nov 2015

- Designed and retrofitted a 2-stroke gasoline engine to a zero-emission compressed air engine in SolidWorks & achieved 0.89 HP at 1886 rpm with 3.11% efficiency, validated via thermodynamic simulation in ANSYS.

3D Analysis of the AGARD 445.6 Wing by FEA for Vibration & Flutter Prediction Oct 2024- Nov 2024

- Designed and analyzed the AGARD 445.6 wing in SolidWorks and ANSYS, performing meshing, determining five modal frequencies under aerodynamic forces, and simulating modal waves for fatigue failure prediction.

Simulation of Metal Forming to Analyze the Effects of Geometry and Friction Using Simufact

- Simulated metal forming in Simufact to study how height-to-diameter ratios and friction affect material flow, barreling, force-displacement, and stress-strain behavior for optimized forming parameters.